Do time machines make physics go crazy? Do they make it impossible to predict how things evolve? If not, then how do the laws of physics choose which trajectory, out of the infinite allowed set, a billiard ball will follow?

In search of an answer, Gunnar Klinkhammer and I in 1989 turned from the *classical* laws of physics to the *quantum* laws. Why the quantum laws? Because they are the Ultimate Rulers of our Universe.

For example, the laws of quantum gravity have ultimate control over gravitation and the structure of space and time. Einstein's classical, general relativistic laws of gravity are mere approximations to the quantum gravity laws—approximations with excellent accuracy when one is far from all singularities and looks at spacetime on scales far larger than 10^{-33} centimeter, but approximations nevertheless (Chapter 13).

Similarly, the classical laws of billiard ball physics, which my students and I had used in studying Polchinski's paradox, are mere approximations to the quantum mechanical laws. Since the classical laws seem to predict "nonsense" (an infinity of possible billiard ball trajectories), Klinkhammer and I turned to the quantum mechanical laws for deeper understanding.

The "rules of the game" are very different in quantum physics than in classical physics. When one provides the classical laws with initial conditions, they predict what will happen afterward (for example, what trajectory a ball will follow); and, if there are no time machines, their predictions are unique. The quantum laws, by contrast, predict only *probabilities* for what will happen, not certainties (for example, the probability that a ball will travel through this, that, or another region of space).

In light of these rules of the quantum mechanical game, the answer that Klinkhammer and I got from the quantum mechanical laws is not surprising. We learned that, if the ball starts out moving along Polchinski's paradoxical trajectory (Figures 14.9 and 14.10 at time t = 3:00 P.M.), then there will be a certain quantum mechanical probability—say, 48 percent—for it subsequently to follow trajectory Ca) in Figure 14.10, and a certain probability-say, also 48 percent—for trajectory (b), and a certain (far smaller) probability for each of the infinity of other classically allowed trajectories. In anyone "experiment," the ball will follow just one of the